

Socioeconomic Deprivation in Ireland: A Multivariate Statistical Analysis| Academic Project

1. Introduction and Research Objective

Socioeconomic deprivation is influenced by a range of factors rather than one single measure. Employment, education, housing, household structure and population characteristics can all play a role in the socioeconomic position of an area. In this project, multivariate statistical methods are used to examine how these factors relate to deprivation across different areas in Ireland.

The analysis focuses on a number of related questions. First, we look at whether areas naturally form different socioeconomic groups based on their characteristics alone. We then examine how stable these groups are and how closely they match the deprivation classifications already given in the Pobal dataset. Principal component analysis is then used to reduce the number of variables and identify the main patterns in the data. Finally, we look at whether these socioeconomic characteristics can be used to classify areas as above or below the deprivation threshold and to predict their deprivation score.

Overall, the project moves from exploring the data to identifying groups, reducing the number of variables and then using these variables for classification and prediction.

2. Data and Sample Construction

The analysis uses the 2022 Pobal Deprivation Index dataset. To obtain a manageable working dataset while retaining a broad cross-section of areas, 1,000 observations are sampled without replacement. A fixed random seed is used so that the analysis is reproducible.

```
set.seed(22450386) # For reproducibility

full_data <- read.csv("pobal_di_2022.csv")

# Randomly sample 1,000 observations without replacement
data <- full_data[sample(nrow(full_data), size = 1000, replace = FALSE),]
```

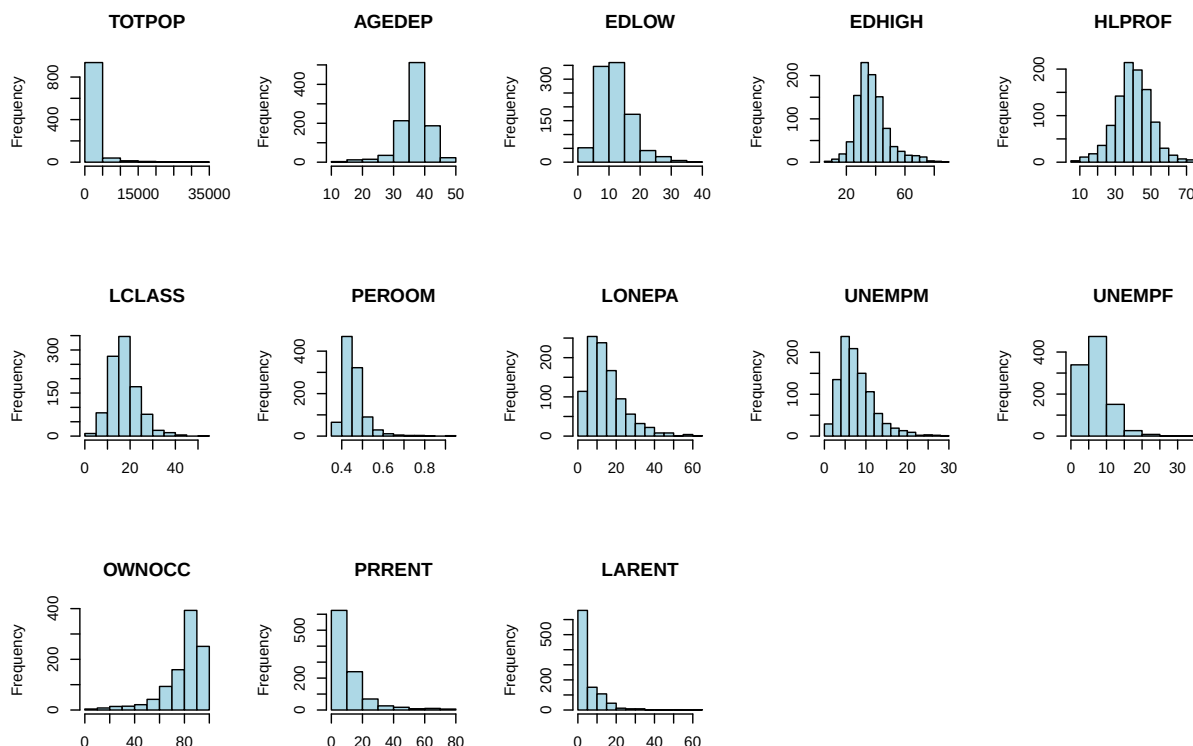
The principal variables of interest are the socioeconomic covariates contained in columns 3–15. These variables describe different characteristics of the areas and will form the basis of the multivariate analysis.

3. Exploratory Analysis of Socioeconomic Characteristics

We first look at the distributions and scales of the individual socioeconomic variables. As there are a large number of variables in the dataset, a full pair plot would be difficult to read. Instead, histograms are used to give a clearer view of how each variable is distributed.

```
par(mfrow = c(3,5))

for(i in 3:15){
  hist(data[,i],
        main = names(data)[i],
        col = "lightblue",
        xlab = "")
}
```



Several variables, including LARENT, PRRENT, UNEMPF and PEROOM, are right-skewed. This indicates that most areas have relatively low values while a smaller number of areas take substantially higher values. Other variables, such as EDHIGH and HLPROF, have a more concentrated central peak and appear closer to a symmetric distribution.

A particularly important feature is the scale of TOTPOP. Total population reaches values close to 20,000, whereas many of the remaining variables generally lie between approximately 0 and 40–60. This difference matters for distance-based methods: without scaling, variables measured on larger numerical scales could dominate Euclidean distance calculations even if they are not intrinsically more important. The socioeconomic variables are therefore rescaled before clustering.

4. Do Irish Areas Form Natural Socioeconomic Groups?

The first part of the analysis looks at whether areas can be grouped naturally based on their socioeconomic characteristics. The deprivation categories are not used when creating these groups. Instead, the clusters are formed using the socioeconomic variables and are then compared with the existing deprivation classifications to see how closely they match.

4.1 Preparing the variables for distance-based clustering

The socioeconomic variables are isolated and rescaled by their standard deviations so that variables with large numerical ranges do not dominate the distance calculations.

```
x <- data[,3:15]

StDev <- apply(x, 2, sd)

# Rescale each variable by its standard deviation
stdX <- sweep(x, 2, StDev, "/")
```

4.2 Hierarchical clustering: identifying a possible group structure

Hierarchical clustering is first used to explore the structure of the observations without fixing the groups in advance. Euclidean distance is used to measure dissimilarity between areas, and Ward's linkage is used to merge clusters.

Ward's method was chosen after considering alternative linkage rules. Single linkage produced a chaining effect, where observations could be joined through sequences of close neighbours, resulting in poorly separated groups. Average linkage also produced less distinct groupings. Ward's method instead chooses merges that minimise the increase in within-cluster variation, producing comparatively compact and interpretable clusters.

```
suppressPackageStartupMessages(library(dendextend))

# Euclidean distances between areas
dist.eucl <- dist(stdX, method = "euclidean")

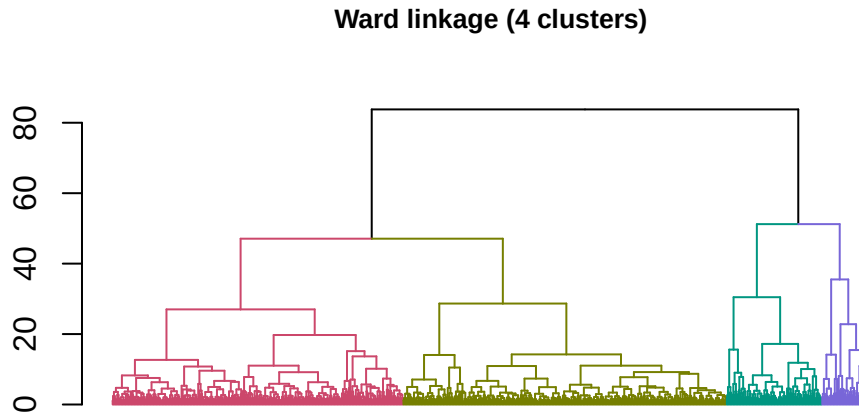
# Ward hierarchical clustering
cl.ward <- hclust(dist.eucl, method = "ward.D2")

n.cl <- 4

cl.ward.dend <- as.dendrogram(cl.ward)
cl.ward.dend <- set(cl.ward.dend, "branches_k_color", k = n.cl)
```

Loading required namespace: colorspace

```
plot(cl.ward.dend,
     leaflab = "none",
     main = "Ward linkage (4 clusters)",
     cex.main = 0.8)
```



```
hcl <- cutree(cl.ward, k = 4)
```

The dendrogram provides evidence about an appropriate number of clusters. Large increases in merge height indicate that relatively dissimilar groups are being combined. Clear jumps are visible around heights of approximately 45 and 85. Cutting the dendrogram below the merge near 45 leaves four substantial branches, motivating a four-cluster representation.

4.3 Do the discovered groups correspond to measured deprivation?

The hierarchical clusters can now be compared with the Pobal deprivation categories. Importantly, these labels were not used to generate the clusters. The comparison therefore assesses whether the patterns discovered from the socioeconomic variables correspond to meaningful differences in deprivation.

```
knitr::kable(
  table(hcl, data$index_lab),
  caption = "Comparison of Hierarchical Clusters with Deprivation Index Labels",
  format = "markdown"
)
```

Table 1: Comparison of Hierarchical Clusters with Deprivation Index Labels

Affluent	Disadvantaged	Marginally Above Average	Marginally Below Average	Very Disadvantaged
0	27	73	283	1
32	0	338	57	0
0	37	1	78	10
9	0	47	7	0

The resulting table suggests that the clustering captures a socioeconomic ordering. Cluster 2 mainly contains affluent and marginally above-average areas, while Cluster 3 contains a large concentration of disadvantaged and very disadvantaged areas. Cluster 1 is dominated by marginally below-average areas, while Cluster 4 contains many affluent and above-average areas. The clusters therefore appear to reflect meaningful socioeconomic differences rather than arbitrary numerical groupings.

4.4 Does k-means reveal a similar structure?

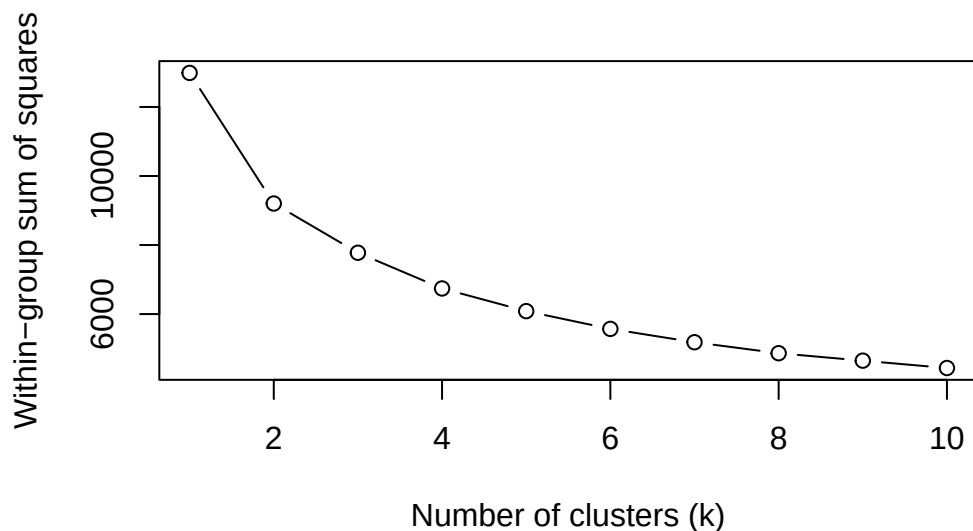
A second clustering method is used to assess whether a comparable structure emerges under a different clustering approach. K-means partitions the observations around cluster centres and seeks to minimise within-cluster variation.

Rather than selecting the number of clusters arbitrarily, the within-group sum of squares is calculated for values of (k) from 1 to 10. The aim is to identify an elbow in the curve, beyond which additional clusters provide comparatively small reductions in within-cluster variation.

```
WGSS <- rep(0,10)

for(k in 1:10){
  WGSS[k] <- sum(kmeans(stdX, centers = k, nstart = 25)$withinss)
}

plot(1:10, WGSS,
     type = "b",
     xlab = "Number of clusters (k)",
     ylab = "Within-group sum of squares")
```



The elbow plot suggests a three-cluster solution. K-means is therefore fitted using three centres, with multiple random initialisations used to reduce sensitivity to the initial starting points.

```
km3 <- kmeans(stdX, centers = 3, nstart = 25)

knitr::kable(
  table(km3$cluster, data$index_lab),
  caption = "K-means Clusters and Deprivation Index Labels",
  format = "markdown"
)
```

Table 2: K-means Clusters and Deprivation Index Labels

Affluent	Disadvantaged	Marginally Above Average	Marginally Below Average	Very Disadvantaged
0	56	2	107	11
31	8	415	305	0
10	0	42	13	0

The three clusters broadly distinguish lower, middle and higher socioeconomic conditions. One cluster is dominated by marginally below-average and disadvantaged areas, another contains a mixture of areas around the middle of the deprivation distribution, and the third is concentrated among marginally above-average and affluent areas. The broad ordering is therefore sensible, although the middle cluster is relatively heterogeneous and the solution provides less detail than the four-group hierarchical clustering result.

4.5 Comparing the two clustering approaches

To quantify the correspondence between the cluster assignments and the deprivation labels, the Adjusted Rand Index (ARI) is calculated. The ARI measures agreement between two classifications while adjusting for agreement that could occur by chance.

```
suppressPackageStartupMessages(library(mclust))

ari_hcl <- adjustedRandIndex(hcl, data$index_lab)
ari_km  <- adjustedRandIndex(km3$cluster, data$index_lab)

results <- data.frame(
  Method = c("Hierarchical (Ward)", "K-means"),
  ARI = c(round(ari_hcl,3), round(ari_km,3))
)

knitr::kable(
  results,
  caption = "Agreement with Deprivation Labels Using the Adjusted Rand Index",
  format = "markdown"
)
```

Table 3: Agreement with Deprivation Labels Using the Adjusted Rand Index

Method	ARI
Hierarchical (Ward)	0.322
K-means	0.153

The hierarchical clustering has an ARI of 0.322 compared with 0.153 for k-means. This means that the hierarchical clusters are more similar to the existing deprivation categories in this sample. However, this does not necessarily mean that hierarchical clustering would perform better on new data.

5. Are the K-means Groups Stable Across Samples?

A clustering result is more convincing if similar group structure can be recovered from different subsets of the data. We therefore investigate the stability of the three-cluster k-means solution using a training and test split.

Half of the observations are used to estimate three k-means cluster centres. Each test observation is then allocated to its nearest training centre. Separately, k-means is fitted directly to the test observations. The two test-set classifications can then be compared using the ARI.

```

library(e1071)

train <- sample(1:nrow(stdX), nrow(stdX)/2)
test <- -train

X.train <- stdX[train, ]
X.test  <- stdX[test, ]

km.train <- kmeans(X.train, centers = 3, nstart = 25)
km.test  <- kmeans(X.test, centers = 3, nstart = 25)

centers <- km.train$centers

# Distances between test observations and training cluster centres
dmat <- as.matrix(dist(rbind(centers, X.test)))[1:3, -(1:3)]

# Assign each test observation to its nearest training centre
S1 <- apply(dmat, 2, which.min)

# Independently estimated cluster membership in the test sample
S2 <- km.test$cluster

ari <- adjustedRandIndex(S1, S2)
round(ari, 3)

```

```
[1] 0.855
```

The ARI of 0.855 indicates a high level of agreement between the two test-set classifications after adjustment for chance. This suggests that the broad three-group k-means structure is relatively stable across the sample split.

6. Can Socioeconomic Characteristics Classify Areas Relative to the Deprivation Threshold?

Having explored socioeconomic structure without using a response variable, the analysis now moves to supervised classification. The continuous relative deprivation index is converted into a binary outcome indicating whether an area's index is at or above zero.

```

suppressPackageStartupMessages(library(MASS))
suppressPackageStartupMessages(library(dplyr))

```

```
data <- data |>
  mutate(is_above_dep = if_else(index_rel >= 0, 1, 0))
```

This binary outcome allows us to ask whether the socioeconomic covariates can distinguish between areas on opposite sides of the deprivation threshold.

6.1 Linear discriminant analysis

Linear Discriminant Analysis (LDA) is first used with all socioeconomic variables as predictors. Cross-validation is used so that each observation is classified without relying on a model fitted using that same observation.

```
lda.cv <- lda(
  is_above_dep ~ .,
  data = data[, c(3:15, which(names(data) == "is_above_dep"))],
  CV = TRUE
)

table(lda.cv$class, data$is_above_dep)
```

```
      0    1
0 459  28
1  41 472
```

```
1 - sum(diag(table(lda.cv$class, data$is_above_dep))) /
  sum(table(lda.cv$class, data$is_above_dep))
```

```
[1] 0.069
```

The misclassification rate is 6.9%, meaning that only a small proportion of observations are incorrectly classified as above or below the deprivation level. This low error rate shows that the LDA model performs well in distinguishing between the two groups.

6.2 Does allowing more flexible class covariance improve classification?

LDA assumes a common covariance matrix across the two classes. Quadratic Discriminant Analysis (QDA) relaxes this assumption by estimating a separate covariance matrix for each group. This provides a more flexible decision boundary but also requires more parameters to be estimated.

```
qda.res.cv <- qda(
  is_above_dep ~ .,
  data = data[, c(3:15, which(names(data) == "is_above_dep"))],
  CV = TRUE
)

table(qda.res.cv$class, data$is_above_dep)
```

```
      0   1
0 435  56
1  65 444
```

```
1 - sum(diag(table(qda.res.cv$class, data$is_above_dep))) /
  sum(table(qda.res.cv$class, data$is_above_dep))
```

```
[1] 0.121
```

The QDA misclassification rate is 12.1%, which is higher than the 6.9% obtained using LDA. In this dataset, allowing each group to have a separate covariance structure does not improve classification. The simpler LDA specification therefore provides the better cross-validated classification performance.

7. What Are the Main Dimensions of Socioeconomic Variation?

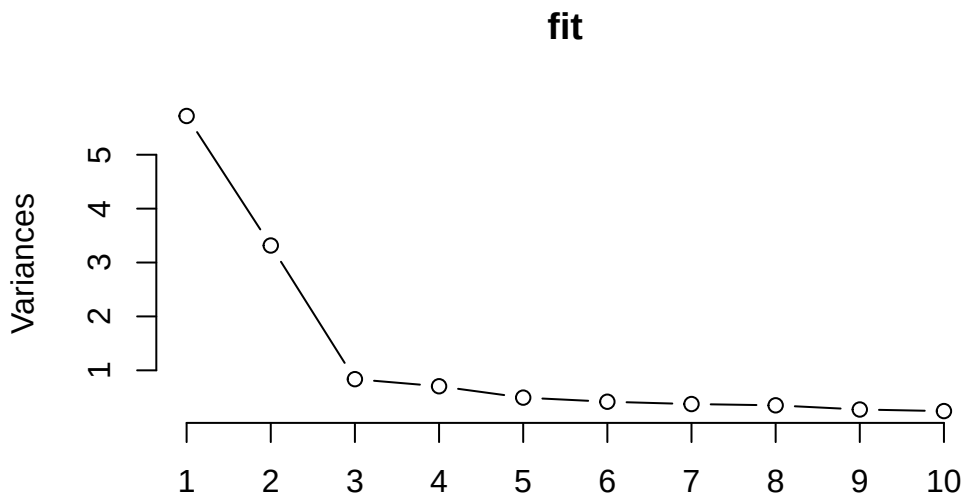
Many of these socioeconomic variables may be correlated and may reflect common underlying socioeconomic dimensions. We use Principal Component Analysis (PCA) to investigate whether the information in the original variables can be represented using a smaller number of uncorrelated components.

```
fit <- prcomp(data[,3:15], scale. = TRUE)
summary(fit)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.3914	1.8209	0.91447	0.83936	0.70258	0.64612	0.6118
Proportion of Variance	0.4399	0.2551	0.06433	0.05419	0.03797	0.03211	0.0288
Cumulative Proportion	0.4399	0.6950	0.75929	0.81349	0.85146	0.88357	0.9124
	PC8	PC9	PC10	PC11	PC12	PC13	
Standard deviation	0.59226	0.52278	0.49372	0.42950	0.2816	0.08767	
Proportion of Variance	0.02698	0.02102	0.01875	0.01419	0.0061	0.00059	
Cumulative Proportion	0.93935	0.96037	0.97912	0.99331	0.9994	1.00000	

```
plot(fit, type = "l")
```



The scree plot shows a substantial decline in explained variance after the first components. From the PCA summary, the first two principal components explain approximately 69.5% of the total variance in the socioeconomic variables. Two components therefore provide a useful low-dimensional summary, while still retaining a substantial majority of the variation present in the original predictors.

7.1 Interpreting the first two principal components

The component loadings show how strongly each original socioeconomic variable contributes to each principal component. Examining these loadings helps move beyond dimensionality reduction and gives the components substantive socioeconomic meaning.

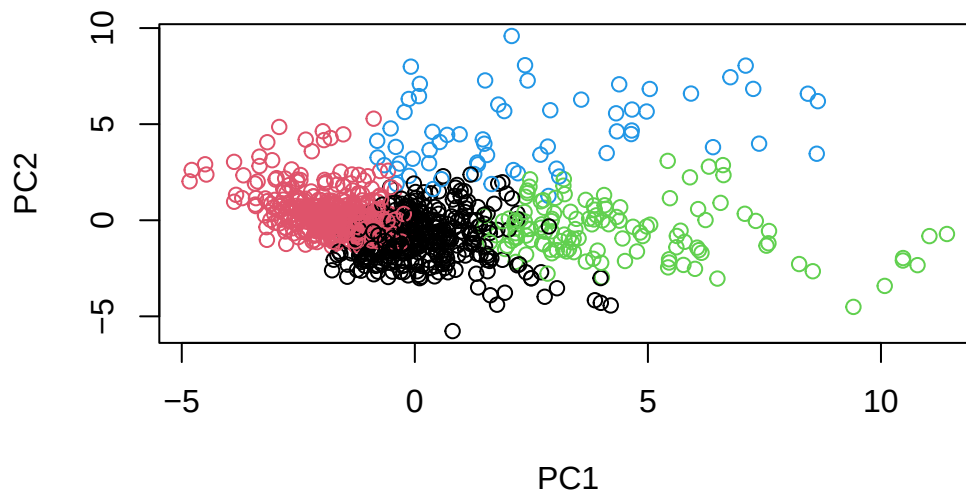
```
loading_matrix <- round(fit$rotation, 2)
loading_matrix[,1:2]
```

	PC1	PC2
TOTPOP	0.12	0.25
AGEDEP	-0.16	-0.38
EDLOW	0.20	-0.39
EDHIGH	-0.17	0.46
HLPROF	-0.34	0.21
LCLASS	0.33	-0.23
PEROOM	0.24	0.28
LONEPA	0.35	0.05
UNEMPM	0.32	-0.10
UNEMPF	0.31	-0.06
OWNOCC	-0.33	-0.28
PRRENT	0.21	0.40
LARENT	0.35	-0.01

```
scores <- predict(fit)

plot(scores[,1:2],
      col = hcl,
      main = "Socioeconomic Areas in the First Two Principal Components")
```

Socioeconomic Areas in the First Two Principal Components



PC1 is strongly related to variables including unemployment, lone-parent households and renting, while PC2 is more strongly associated with education and housing characteristics. These loadings suggest that the components represent broader latent dimensions of socioeconomic conditions rather than isolated individual indicators.

Colouring the observations according to the hierarchical cluster assignments provides a visual assessment of how much of the previously identified group structure is represented by the first two principal components. There is visible separation between some groups, although there is also overlap. This indicates that the first two components capture an important part of the socioeconomic structure but do not provide a complete representation of all distinctions between areas.

8. How Much Cluster Structure Is Preserved After Dimensionality Reduction?

The two principal components retain ~69.5% of the total predictor variation, so its useful to investigate whether clustering in this reduced space produces similar groupings to clustering on the full set of socioeconomic variables.

Hierarchical clustering is therefore repeated using only PC1 and PC2.

```
pca_data <- scores[,1:2]

dist_pca <- dist(pca_data, method = "euclidean")

cl_pca <- hclust(dist_pca, method = "ward.D2")
hcl_pca <- cutree(cl_pca, k = 4)

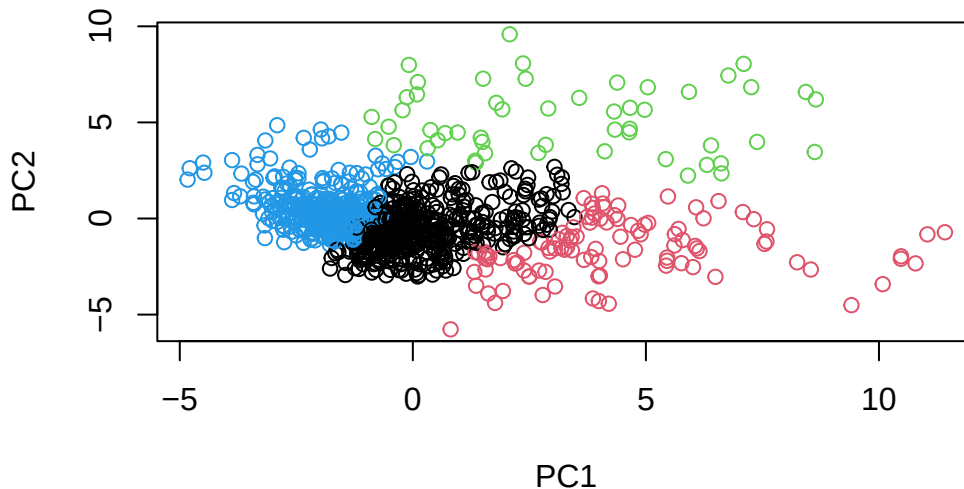
knitr::kable(
  table(hcl, hcl_pca),
  caption = "Full-space and PCA-space Hierarchical Cluster Assignments",
  format = "markdown"
)
```

Table 4: Full-space and PCA-space Hierarchical Cluster Assignments

	1	2	3	4
1	350	27	0	7
2	53	0	1	373
3	43	78	5	0
4	11	0	46	6

```
plot(scores[,1], scores[,2],
     col = hcl_pca,
     xlab = "PC1",
     ylab = "PC2",
     main = "Hierarchical Clustering in the First Two Principal Components")
```

Hierarchical Clustering in the First Two Principal Components



As the cluster numbers are just labels, comparing them directly can give the wrong impression if the same groups are given different numbers. The Adjusted Rand Index avoids this problem and gives a better measure of how similar the two clustering results are.

```
ari_pca_clusters <- adjustedRandIndex(hcl, hcl_pca)
round(ari_pca_clusters, 3)
```

```
[1] 0.63
```

This comparison looks at whether PC1 and PC2 keep enough information to produce similar clusters to the original data. Even though the first two principal components explain a large amount of the variation, some information that is useful for forming clusters may still be lost.

8.1 Benefits and limitations of clustering in a reduced-dimensional space

Clustering after PCA makes the data easier to work with because the 13 socioeconomic variables are reduced to two principal components. This also makes it possible to plot the observations and clusters in two dimensions.

However, using only the first two components means that some information is lost. In this case, PC1 and PC2 explain about 69.5% of the total variation, so around 30.5% is left in the remaining components. If some of this missing variation is important for separating groups, the clusters may differ from those formed using all of the original variables.

This means clustering after PCA is useful for simplifying and visualising the data.

9. Can Two Principal Components Retain Enough Information for Deprivation Classification?

The PCA results suggest that much of the socioeconomic variation can be summarised by PC1 and PC2. We can therefore ask a stronger question: are these two components alone sufficient to classify areas relative to the deprivation threshold?

LDA is refitted using only PC1 and PC2 as predictors.

```
lda_pca_data <- data.frame(  
  PC1 = scores[,1],  
  PC2 = scores[,2],  
  is_above_dep = data$is_above_dep  
)  
  
lda.PCA <- lda(is_above_dep ~ PC1 + PC2, data = lda_pca_data)  
  
lda_PCA_cv <- lda(  
  is_above_dep ~ PC1 + PC2,  
  data = lda_pca_data,  
  CV = TRUE  
)  
  
tab <- table(lda_PCA_cv$class, lda_pca_data$is_above_dep)  
  
1 - sum(diag(tab)) / sum(tab)
```

```
[1] 0.096
```

The misclassification rate is 9.6%. This is only moderately higher than the 6.9% obtained when LDA used all 13 socioeconomic variables. The result therefore shows that a large amount of the information relevant for deprivation classification is concentrated in the first two principal components, although some predictive information is lost through dimensionality reduction.

9.1 Visualising the classification rule

Because the model now has only two predictors, its classification rule can be displayed directly in the PC1–PC2 plane.

```

boundary <- function(model, data, class = NULL, predict_type = "class",
                      resolution = 100, showgrid = TRUE, ...) {

  if(!is.null(class)) cl <- data[,class] else cl <- 1
  data <- data[,1:2]
  k <- length(unique(cl))

  plot(data, col = as.integer(cl)+1L, pch = as.integer(cl)+1L, ...)

  r <- sapply(data, range, na.rm = TRUE)
  xs <- seq(r[1,1], r[2,1], length.out = resolution)
  ys <- seq(r[1,2], r[2,2], length.out = resolution)
  g <- cbind(rep(xs, each=resolution), rep(ys, time = resolution))
  colnames(g) <- colnames(r)
  g <- as.data.frame(g)

  p <- predict(model, g, type = predict_type)
  if(is.list(p)) p <- p$class
  p <- as.factor(p)

  if(showgrid) points(g, col = as.integer(p)+1L, pch = ".")

  z <- matrix(as.integer(p), nrow = resolution, byrow = TRUE)
  contour(xs, ys, z, add = TRUE, drawlabels = FALSE,
          lwd = 2, levels = (1:(k-1))+.5)

  invisible(z)
}

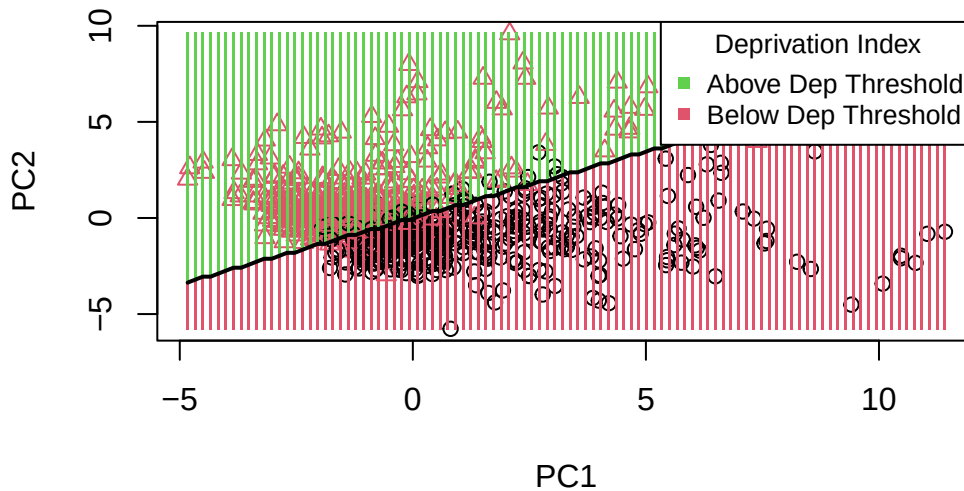
boundary(
  lda.PCA,
  lda_pca_data,
  class = "is_above_dep",
  main = "LDA Decision Boundary Using PC1 and PC2"
)

legend("topright",
      legend = c("Above Dep Threshold", "Below Dep Threshold"),
      col = c(3, 2),
      pch = 15,
      title = "Deprivation Index",
      cex = 0.8,

```

```
bg = "white",
box.col = "black")
```

LDA Decision Boundary Using PC1 and PC2



The straight decision boundary comes from the assumptions of LDA. Most observations are on the correct side of the boundary, which matches the low misclassification rate. The plot also shows that PC1 and PC2 are useful for separating areas that are above or below the deprivation threshold.

10. Can the Underlying Socioeconomic Dimensions Predict the Continuous Deprivation Index?

The final part of the analysis looks at whether the actual deprivation score, `index_rel`, can be predicted using the socioeconomic variables rather than only classifying areas into two groups.

Principal Component Regression (PCR) is used for this. PCR first reduces the original variables into principal components and then uses these components in the regression model. This is useful when the original variables are correlated and also allows us to control how many components are included in the model.

10.1 Fitting the PCR model

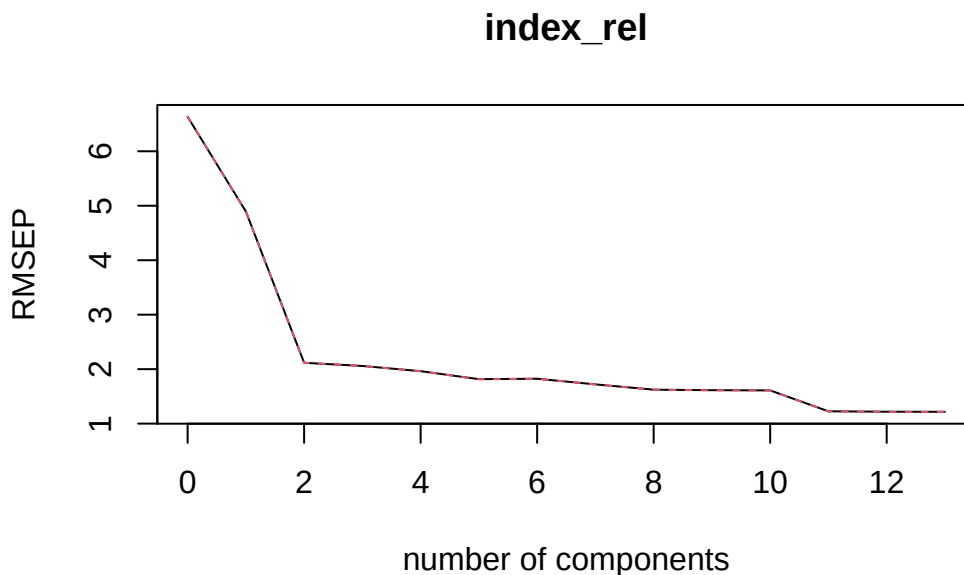
```
suppressPackageStartupMessages(library(pls))

pcr.fit <- pcr(
  index_rel ~ .,
  data = data[, c(3:15, which(names(data) == "index_rel"))],
  scale = TRUE,
  validation = "CV"
)
```

The predictors are scaled before constructing the components so that variables measured on larger numerical scales do not dominate the PCA step. Cross-validation is incorporated into the model to assess predictive performance and help determine how many components should be retained.

10.2 How many components are needed to predict deprivation?

```
validationplot(pcr.fit, val.type = "RMSEP")
```



```
summary(pcr.fit)
```

```
Data:   X dimension: 1000 13
       Y dimension: 1000 1
```

Fit method: svdpc
Number of components considered: 13

VALIDATION: RMSEP

Cross-validated using 10 random segments.

	(Intercept)	1 comps	2 comps	3 comps	4 comps	5 comps	6 comps
CV	6.633	4.895	2.117	2.059	1.963	1.816	1.824
adjCV	6.633	4.894	2.116	2.059	1.962	1.814	1.824
	7 comps	8 comps	9 comps	10 comps	11 comps	12 comps	13 comps
CV	1.720	1.624	1.614	1.610	1.226	1.219	1.217
adjCV	1.719	1.622	1.612	1.609	1.224	1.217	1.215

TRAINING: % variance explained

	1 comps	2 comps	3 comps	4 comps	5 comps	6 comps	7 comps
X	43.99	69.50	75.93	81.35	85.15	88.36	91.24
index_rel	45.72	89.87	90.50	91.37	92.61	92.61	93.45
	8 comps	9 comps	10 comps	11 comps	12 comps	13 comps	
X	93.93	96.04	97.91	99.33	99.94	100.00	
index_rel	94.18	94.31	94.38	96.75	96.80	96.81	

The Root Mean Squared Error of Prediction (RMSEP) measures how far the predicted values are from the actual values, with a lower RMSEP indicating better predictions. The largest improvement occurs when the first two principal components are included. Adding further components only leads to small improvements.

The first two components explain around 69.5% of the variation in the socioeconomic variables and 89.87% of the variation in the response explained by the PCR model. This shows that the first two components contain a large amount of the information needed to predict the deprivation score.

This is similar to what we found earlier using PCA and LDA. PC1 and PC2 capture much of the variation in the socioeconomic variables and are also useful for both classifying and predicting deprivation.

10.3 Strengths and limitations of PCR in this setting

One advantage of PCR is that it reduces a large number of correlated variables to a smaller number of uncorrelated principal components. This simplifies the regression model and can help to reduce problems caused by highly correlated predictors.

However, PCA creates the components based on the variation in the predictor variables without considering the deprivation score. Therefore, a component that explains less variation in the socioeconomic variables could still be useful for predicting deprivation.

Another limitation is that the principal components are made up of combinations of several variables. This makes the results harder to interpret because we cannot directly identify the effect of one individual socioeconomic variable from the PCR coefficients.

11. Overall Discussion and Conclusion

The results show that there are clear relationships between the socioeconomic variables and deprivation across the areas in the sample.

The clustering analysis showed that areas can be grouped based only on their socioeconomic characteristics. Hierarchical clustering had a stronger relationship with the existing Pobal deprivation categories than k-means. However, the train and test analysis also showed that the k-means clusters were quite stable across different samples of the data.

PCA showed that the 13 socioeconomic variables can be reduced to a much smaller number of components while keeping a large amount of the information in the data. PC1 and PC2 alone explain around 69.5% of the total variation. The variables contributing to these components include unemployment, lone-parent households, renting, education and housing, showing that several socioeconomic characteristics are related to the main patterns in the data.

The classification results also showed a strong relationship between these characteristics and deprivation. LDA using all of the socioeconomic variables had a misclassification rate of 6.9%. When only PC1 and PC2 were used, this increased to 9.6%. This shows that the first two components contain much of the information needed to classify areas, although some useful information is lost by reducing the data to only two components. QDA had a higher misclassification rate than LDA, so the additional complexity of QDA did not improve the results.

Finally, PCR showed that the principal components can also be used to predict the continuous deprivation score. Overall, the results suggest that deprivation is related to several socioeconomic characteristics rather than any one variable. These characteristics can be used to group areas and much of their information can be reduced to a small number of components which are useful for both classifying and predicting deprivation.